

Understanding and Applying DevOps Principles

Lab 2

Team details

- Team name: not provided
- Team number: not provided
- Team members
 - Qixuan Yang (20719443)
 - Ronghao Hu (20719451)
 - Hanbo Zhao (20719456)

Background of the organization

- About 100 people at one site; three Dev teams of 10-15 people; three web apps share a database. Teams use their own processes/tools. Dev tests functions; Test checks integration/acceptance manually; Ops deploys to Google Cloud. Sales collects needs; communication is mainly informal email.

Task 1 - Identify list of challenges

- Communication: lost priorities and decisions. Business/data: availability needs and shared-schema risks. Delivery: manual test queues and Dev-Test-Ops handoffs. Learning: possible knowledge silos and conflicting incentives. Each is a risk to investigate, not a confirmed failure.

Task 1 - Challenge 1 - Communication

- Assumptions
 - A1: no shared backlog or decision log; incentives may differ between Dev and Ops.
- Implication
 - Priorities and context may be lost; conflicting goals discourage sharing. Agree one backlog, named owners and shared outcomes.

Task 1 - Challenge 2 – Business area

- Assumptions
 - A2: assume customer-facing subscription services need reliable releases.
 - A3: a shared-schema change may break another application.
- Implication
 - Availability can affect revenue; schema compatibility affects all three apps. Use staged releases and versioned, compatible migrations.

Task 1 - Challenge 3 – Release queues

- Assumptions
 - A4: manual tests and deployment knowledge have limited capacity.
- Implication
- Queues delay releases; larger batches increase rework. Automate critical checks, limit WIP and share release ownership.

Task 2 - Plan - Communication

- Name : Shared backlog and pilot squad
- Purpose, actions, owner and timeline
 - W1-2: lead + Sales map queues and baseline metrics. Dev/Test/Ops form a pilot squad; Sales owns priorities. Use one board, WIP ≤ 3 and shared done/release rules. Continue development; measure waiting weekly. Goal: fewer handoffs and lost decisions.

Task 2 - Plan – Business Area

- Name : Safe delivery pilot
- Purpose, actions, owner and timeline
 - W3-5: Dev/Test automate build, unit and all-app integration checks. W4-6: Ops tests staging, compatible migrations and rollback. Gate production release; monitor health. W7-8: review pilot evidence. Lead scales to other teams in W9-12 only if gates pass.

Task 3 – Continuous Improvement - **Evaluate the plans you have made so far considering the third way:**

- Boards and tests support learning; rigid gates and personal metric rankings may discourage experiments.
- Protect learning time, delegate staging trials, and review failures without blame; keep production release checks.

Task 3 – Continuous Improvement – **Suggest new items to the plan**

- From W2, squad lead reserves 10% of pilot time for a logged experiment every two weeks: hypothesis, baseline, staging trial, success/stop rule and decision. From W3, Ops/Test review incidents or near misses within two working days and assign corrective actions. Dev/Test run weekly cross-role demos and a monthly all-team session; publish lessons in a shared wiki and verify another team reuses them.

Task 3 – Continuous Improvement – **Suggest improvement for the complete plan**

- W1-2: baseline, shared backlog and trust charter. W3-5: CI/testing plus pairing. W6-8: small releases, monitoring and experiments. W9-12: scale after gates; share lessons. Lead protects 10% learning time; CEO reviews delivery and reliability.